



BIOSOLIDS LAND APPLICATION SITE IDENTIFICATION FORM

*Authorized under 1994 PA 451, as amended. Completion of form is required.
Applicant may be subject to civil and/or criminal penalties for providing false information.*

Complete the following information and provide the attachments listed on the second page for each new or updated biosolids land application site. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e., new landowner).

For additional information, Biosolids Program Staff are listed at: https://www.michigan.gov/documents/deq/wrd-biosolids-staff_402800_7.pdf.

Facility Name		NPDES Permit or COC Number		Request: New Site / Site Update / Site transfer	
Site ID Name (ex T/R/S – Owner (05N08E03-LB01))			Site location description (street name, nearest crossroad)		
County			Township		
(ex: NW, NE, SW, SE) 1/4		(ex: NW, NE, SW, SE) 1/4		Township (ex: 05N)	Range (ex: 08E)
Latitude (ex: N 42.09651)		Longitude (ex: -81.09651)		Method Used (i.e., GPS, topo)	
Name of Owner				Phone	
Street or P.O. Box			City or Town		State
Street or P.O. Box			City or Town		State
Is there a written agreement from the owner to land apply? ☐ Yes ☐ No (written agreement is required)					
Previous Landowner's Name (if site update)				Property Purchase Date	
Name of Lessee/Farmer (if different)				Phone	
Street or P.O. Box			City or Town		State
Street or P.O. Box			City or Town		State
Site Type ☐ Agricultural ☐ Forest ☐ *Reclamation Site (separate approval is required) ☐ Lawn/Home Garden (requires EQ Biosolids)					
Soils Information Date of last soils analysis _____ *Phosphorous _____ units _____ Potassium _____ units _____ pH _____ Soil type(s) _____ *300 lb/ac (or 150 ppm) Bray P1 maximum limit			Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☐ *Higher (Requires an approved site reclamation plan)		
Expected Crops/Vegetation to be grown Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre					
Total acreage of site		Acreage used for crops		Acreage used for land application	
Does this site have a high potential for public exposure? ☐ Yes ☐ No			If agricultural, is the site tilled? ☐ Yes ☐ No ☐ NA		

CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: John Rydquist Signed: John Rydquist
Address: 9897 S 43 Rd. Date: 8-2-18
Township: Clam Lake Section: 27 Acres: 64
City WWTP Field Number: 21N9W27S R01
If Rented, Farmer: _____
Address: _____

**MICHIGAN STATE UNIVERSITY
SOIL AND PLANT NUTRIENT LABORATORY
EAST LANSING, MICHIGAN 48824-1325
(517) 355-0218**

SOIL TEST REPORT FOR:				CONSULTANT			
CITY OF CADILLAC 200 N. LAKE ST. CADILLAC MI 49601				WEXFORD COUNTY MSUE 401 N. LAKE STREET83 CADILLAC MI 49601 231-779-9480			
DATE	LAB #	COUNTY	Previous Crop	ACRES	FIELD ID	SOIL	TEXTURE
8/17/2020	247701	Wexford		20	21n9w27jr01	Mineral	

SOIL NUTRIENT LEVELS			Below Optimum	Optimum	Above Optimum
¹ Soil pH 6.8	Lime Index				
² Phosphorus (P)	53	ppm	██████████	████████████████	██
³ Potassium (K)	98	ppm	██████████	█████	
³ Magnesium (Mg)	119	ppm	██████████	████████████████	████

ADDITIONAL RESULTS:					Optional Tests:						
Calcium (Ca) (ppm)	CEC (meq/100 g)	% of Exchangeable Bases			Micronutrients (ppm)					Organic Matter %	Nitrate-N ppm
		K	Mg	Ca	B	Cu	Mn	Zn	Fe		
973	6.1	4.1	16.2	79.7							

RECOMMENDATIONS:

Limestone: NONE

Tillage Depth: 9 inches

Target pH = 6.5

% Stand:

Plant Nutrients:					Micronutrient: (Optional)				
Year	Crop	Expected Yield	Nitrogen (lb N/A)	Phosphate (lb P ₂ O ₅ /A)	Potassium (lb K ₂ O/A)	Boron (lb B/A)	Manganese (lb Mn/A)	Zinc (lb Zn/A)	Copper (lb Cu/A)
1	Corn silage	18 ton	135	0	145	0.0			

Use a starter fertilizer high in nitrogen (30-40 lb N/A) when planting into heavy residue.

Use a phosphate containing starter fertilizer (25 to 40 lb P₂O₅) when planting into heavy residue.

Although phosphorus is not required to obtain the projected yield, a phosphorus starter (25 lb P2O5/A) may improve early plant growth.

For questions about interpreting your soil results go to the following website: <http://msue.anr.msu.edu/experts> and contact the MSUE field crops agent in your area.

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231-839-4444

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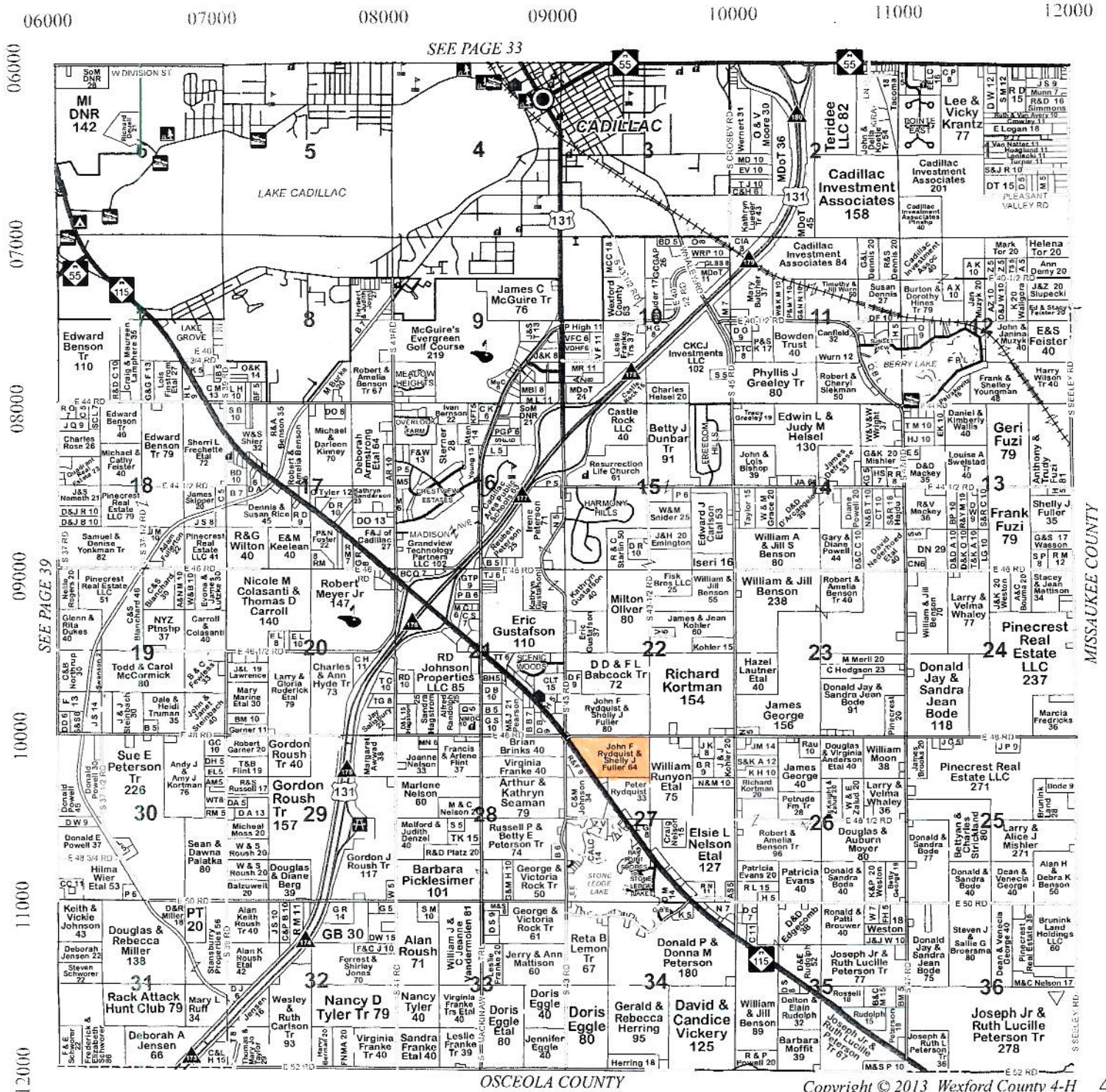
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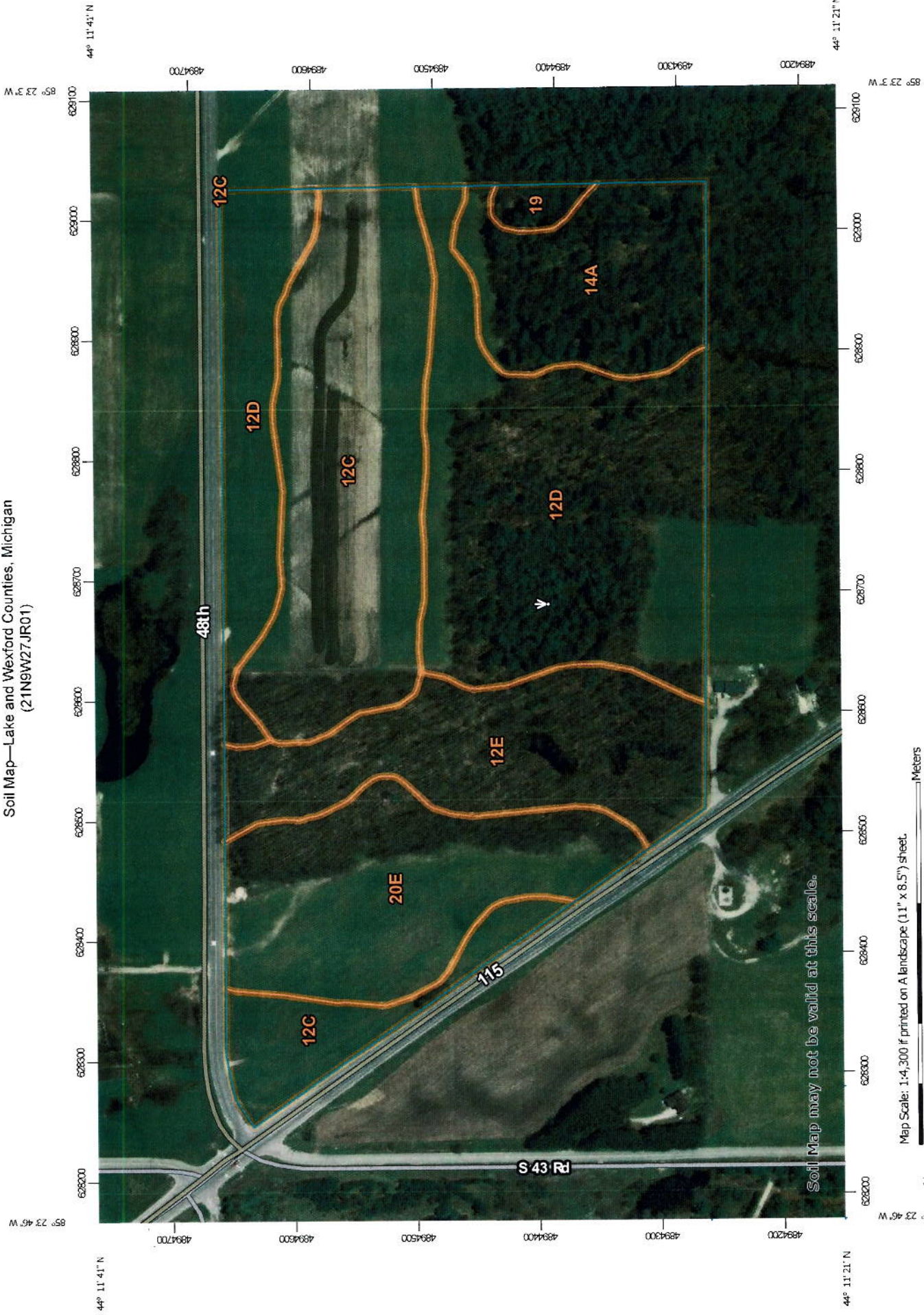
www.kettunencenter.org
(231) 829-3421



CLAM LAKE T21N • R9W



Soil Map—Lake and Wexford Counties, Michigan (21N9W27JR01)



MAP LEGEND

	Area of Interest (AOI)		Spill Area
	Area of Interest (AOI)		Stony Spot
	Soils		Very Stony Spot
	Soil Map Unit Polygons		Wet Spot
	Soil Map Unit Lines		Other
	Soil Map Unit Points		Special Line Features
	Special Point Features		Streams and Canals
	Blowout		Transportation
	Borrow Pit		Rails
	Clay Spot		Interstate Highways
	Closed Depression		US Routes
	Gravel Pit		Major Roads
	Gravelly Spot		Local Roads
	Landfill		Background
	Lava Flow		Aerial Photography
	Marsh or swamp		
	Mine or Quarry		
	Miscellaneous Water		
	Perennial Water		
	Rock Outcrop		
	Saline Spot		
	Sandy Spot		
	Severely Eroded Spot		
	Sinkhole		
	Slide or Slip		
	Sodic Spot		

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL:

Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Lake and Wexford Counties, Michigan
Survey Area Data: Version 11, Oct 6, 2017

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: May 9, 2014—Sep 19, 2016

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
12C	Emmet-Montcalm complex, 6 to 12 percent slopes	16.5	25.7%
12D	Emmet-Montcalm complex, 12 to 18 percent slopes	21.2	33.0%
12E	Emmet-Montcalm complex, 18 to 40 percent slopes	9.3	14.4%
14A	Allendale loamy sand, 0 to 4 percent slopes	6.8	10.5%
19	Lupton muck, 0 to 1 percent slopes	0.6	1.0%
20E	Montcalm-Graycalm complex, 12 to 30 percent slopes	10.0	15.5%
Totals for Area of Interest		64.4	100.0%